UChicago REU Notes Week 2

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Contents

1 Review: Symmetric Monoidal Categories

Definition 1.1. A category $\mathcal{V}$ is a *monoidal category*, if there is a bifunctor $\otimes : \mathcal{V}\times\mathcal{V}\to\mathcal{V}$ (called *monoidal product*), and an object $\kappa\in\mathcal{V}$ (called *unit/monoidal unit*), together with the following natural isomorphisms:

- The *associator* $a : (\_\otimes\_)\otimes\_ \xrightarrow{\sim} \_\otimes(\_\otimes\_)$ of functors $\mathcal{V}\times\mathcal{V}\times\mathcal{V}\xrightarrow{\sim}\mathcal{V}$. (So, $(X\otimes Y)\otimes Z\cong X\otimes(Y\otimes Z)$ in a canoncial way).

- The *left unit* $\lambda : \kappa\otimes\_ \xrightarrow{\sim} \mathrm{Id}_{\mathcal{V}}$ ($\kappa\otimes X\cong X$ in a canonical way).

- The *right unit* $\rho : \_\otimes\kappa \xrightarrow{\sim} \mathrm{Id}_{\mathcal{V}}$ ($X\otimes\kappa\cong X$ in a canonical way).

They also satisfy two extra axioms:

- *Triangle Axiom:*

$$\begin{tikzcd}
(X\otimes\kappa)\otimes Y \arrow[rr, "a_{X,\kappa,Y}"] \arrow[dr, "\rho_X\otimes\mathrm{id}_Y"'] & & X\otimes(\kappa\otimes Y) \arrow[dl, "\mathrm{id}_X\otimes\lambda_Y"] \\
& X\otimes Y &
\end{tikzcd}$$

- *Pentagon Axiom:*

$$\begin{tikzcd}
((X\otimes Y)\otimes Z)\otimes W \arrow[rr, "a_{X\otimes Y,Z,W}"] \arrow[d, "a_{X,Y,Z}\otimes\mathrm{id}_W"'] & & (X\otimes Y)\otimes(Z\otimes W) \arrow[d, "a_{X,Y,Z\otimes W}"] \\
(X\otimes(Y\otimes Z))\otimes W \arrow[dr, "a_{X,Y\otimes Z,W}"'] & & X\otimes(Y\otimes(Z\otimes W)) \\
& X\otimes((Y\otimes Z)\otimes W) \arrow[ur, "\mathrm{id}_X\otimes a_{Y,Z,W}"'] &
\end{tikzcd}$$

Based on *Maclane's Coherence Theorem*, the above two axioms guarantees all finite sequences of a, λ, ρ and their inverses agree, namely there is a unique isomorphism $P_1 \to P_2$, where P_i are (possibly) different orders of performing product on $X_1, ..., X_n \in \mathcal{V}$. For reference, cf. *Category Theory for Working Mathematicians*.

This guarantees all monoidal category to be equivalent to a *strict monoidal category*,namely all a, λ, ρ above are "identity transformations", or $1 \otimes X = X, X \otimes 1 = X$, and $(X \otimes Y) \otimes Z = X \otimes (Y \otimes Z)$, so WLOG one can assume the product $\otimes$ is associative.

Definition 1.2. A monoidal category $\mathcal{V}$ is *braided*, if there is a natural isomorphism $B : \_1 \otimes \_2 \xrightarrow{\sim} \_2 \otimes \_1$ (so $B_{X,Y} : X \otimes Y \xrightarrow{\sim} Y \otimes X$ is an isomorphism), together with the *Hexagon axioms* satisfied:

$$\begin{tikzcd}
(X \otimes Y) \otimes Z \arrow[r, "a_{X,Y,Z}"] \arrow[d, "B_{X,Y} \otimes \mathrm{Id}_Z"'] & X \otimes (Y \otimes Z) \arrow[r, "B_{X,Y \otimes Z}"] & (Y \otimes Z) \otimes X \arrow[d, "a_{Y,Z,X}"] \\
(Y \otimes X) \otimes Z \arrow[r, "a_{Y,X,Z}"'] & Y \otimes (X \otimes Z) \arrow[r, "\mathrm{id}_Y \otimes B_{X,Z}"'] & Y \otimes (Z \otimes X)
\end{tikzcd}$$

$$\begin{tikzcd}
X \otimes (Y \otimes Z) \arrow[d, "\mathrm{id}_X \otimes B_{Y,Z}"'] & (X \otimes Y) \otimes Z \arrow[l, "a_{X,Y,Z}"'] \arrow[r, "B_{X \otimes Y,Z}"] & Z \otimes (X \otimes Y) \\
X \otimes (Z \otimes Y) & (X \otimes Z) \otimes Y \arrow[l, "a_{X,Z,Y}"] \arrow[r, "B_{X,Z} \otimes \mathrm{id}_Y"'] & (Z \otimes X) \otimes Y \arrow[u, "a_{Z,X,Y}"']
\end{tikzcd}$$

$\mathcal{V}$ is a *symmetric monoidal category*, if $B_{Y,X} = B_{X,Y}^{-1}$ for all $X, Y \in \mathcal{V}$.

The two hexagons are uniqueness of 3-cycle permutations on three consecutive objects in a product, hence together with B serving as transpositions, it guarantees unique isomorphism between all permutations of finite products, i.e., it imposes that there is a single way of doing such permutation (so there's no "nontrivial braiding", when $B_{Y,X} \circ B_{X,Y} : X \otimes Y \to Y \otimes X \to X \otimes Y$ is not identity). From my naive understanding of coherence theorem, WLOG we can again treat $X \otimes Y = Y \otimes X$.

Example 1.3. Everyone loves unital commutative ring R, so I use category of R-modules as example :)

Equipped with tensor product $\otimes_R$, and R itself as tensor unit (up to isomorphism), we see $\mathsf{R{-}Mod}$ forms a symmetric monoidal category due to following natural isomorphisms:

$$\begin{cases}
(M \otimes_R N) \otimes_R L \cong M \otimes_R (N \otimes_R L) & , \quad (m \otimes n) \otimes \ell \mapsto m \otimes (n \otimes \ell) \\
R \otimes_R M \cong M & , \quad r \otimes m \mapsto rm \\
M \otimes_R R \cong M & , \quad m \otimes r \mapsto rm \\
M \otimes_R N \cong N \otimes_R M & , \quad m \otimes n \mapsto n \otimes m
\end{cases}$$

Now, observe that $\mathsf{R{-}Mod}$ is self-enriched, meaning $\mathrm{Hom}_R(\_, \_) : \mathsf{R{-}Mod}^{\mathrm{op}} \times \mathsf{R{-}Mod} \to \mathsf{R{-}Mod}$ is a bifunctor (also called an *internal hom functor*). Moreover, there's the tensor-hom adjunction:

$$\mathrm{Hom}_R(M \otimes_R N, L) \cong \mathrm{Hom}_R(M, \mathrm{Hom}_R(N, L)), \quad f \mapsto (m \mapsto f(m \otimes \_))$$

This indicates $(\_ \otimes_R N \dashv \mathrm{Hom}_R(N, \_))$ is an adjoint pair, guaranteeing $\_ \otimes_R N$ commutes with all colimits. For instance, $(M \oplus L) \otimes_R N = (M \otimes_R N) \oplus (L \otimes_R N)$. This gives rise to the following definition:

Definition 1.4. $\mathcal{V}$ is a *symmetric monoidally closed category*, if there exists an *internal hom functor* $\mathcal{V}(\_, \_) : \mathcal{V}^{\mathrm{op}} \times \mathcal{V} \to \mathcal{V}$, such that the adjunction $(\_ \otimes X \dashv \mathcal{V}(X, \_))$ holds for all $X \in \mathcal{V}$.

$\mathcal{V}$ is a *symmetric monoidally cocomplete category*, if it's cocomplete, together with $\_ \otimes X, X \otimes \_$ preserves colimits for all $X \in \mathcal{V}$.

If people knows about Yoneda Lemma, welcome to look into the last section, which stregnthen the isomorphism in monoidal closeness to natural isomorphism $\mathcal{V}(\_ \otimes X, \_) \cong \mathcal{V}(\_, \mathcal{V}(X, \_))$ on the bifunctors $\mathcal{V}^{\mathrm{op}} \times \mathcal{V} \to \mathcal{V}$.

Example 1.5. The category of unital commutative rings CRing with $\otimes_{\mathbb{Z}}$ as monoidal product is in fact not monoidally closed. Suppose the contrary there exists such bifunctor $C : \mathsf{CRing}^{\mathrm{op}} \times \mathsf{CRing} \to \mathsf{CRing}$, such that the adjunction holds, then the following isomorphism holds for any rings $R, S \in \mathsf{CRing}$:

$$\mathrm{Hom}_{\mathsf{CRing}}(R, S) \cong \mathrm{Hom}_{\mathsf{CRing}}(\mathbb{Z} \otimes_{\mathbb{Z}} R, S) \cong \mathrm{Hom}_{\mathsf{CRing}}(\mathbb{Z}, C(R, S)) = \{*\}$$

The final implication follows from the fact $\mathbb{Z} \in \mathsf{CRing}$ is initial. Yet, here's a counterexample: $\mathrm{Hom}_{\mathsf{CRing}}(\mathbb{C}, \mathbb{C})$ as at least two maps, including identity $(z \mapsto z)$ and conjugation $(z \mapsto \overline{z})$, so we reach a contradiction.

Notice this is distinct from CRing being monoidally cocomplete: Coequalizer exists in CRing (by quotienting out the ideal generated by difference of outputs), and $\otimes_{\mathbb{Z}}$ is the coproduct in CRing, hence all colimits exist (cf. *Limit Existence Theorem*). Moreover, colimits commutes with all colimit, in particular so does $\otimes_{\mathbb{Z}}$.

Remark 1.6. For some more abstract examples, we assume $\mathcal{V}$ is monoidally cocomplete. If suitable, also assume internal hom functor exists.

One reason to assume this, is to make sure some of the construction makes sense: In some literature professor May (implicitly) assumes existence of colimits, monoidal product commutes with colimits, and occasionally use internal hom as an example.

Notation 1.7. Let Σ_j denotes the jth symmetric group for all $j \in \mathbb{N}$. In particular, $\Sigma_0 = \Sigma_1 = \{1\}$ as trivial group.

Let Σ denote the category of finite numbers, with objects being $\mathbf{j} := \{1, ..., j\}$, and homset

$$\mathrm{Hom}_{\Sigma}(\mathbf{i}, \mathbf{j}) = \begin{cases} \emptyset & , \quad i \neq j \\ \Sigma_j & , \quad i = j \end{cases}$$

Also, let $\emptyset \in \mathcal{V}$ be the initial object. Since initial object is a colimit, $\emptyset$ exists, moreover $\emptyset \otimes X \cong \emptyset$ for all $X \in \mathcal{V}$ once assume monoidal cocompleteness.

2 Intro: Operads

As an example, we'll start with category of compactly generated based space $\mathcal{T}^*$, with cartesian product $\times$ as monoidal product, singleton $\{*\}$ as monoidal unit.

Example 2.1. Given $X \in \mathcal{T}^*$, define $\mathcal{E}_X(j) := \mathrm{Hom}(X^j, X)$ for each $j \in \mathbb{N}$.

Let natural numbers $j = j_1 + ... + j_k$, each $g_i \in \mathcal{E}_X(j_i)$ defines a map $g_i \circ \pi_i : X^j = \prod_{i=1}^k X^{j_i} \to X$, hence $g_1, ..., g_k$ induces a product map $\Pi g_i : X^j \to X^k$. Now, given any $f \in \mathcal{E}_X(k)$, $f \circ g : X^j \to X^k \to X$ is an element in $\mathcal{E}_X(j)$. This induces the following structure map:

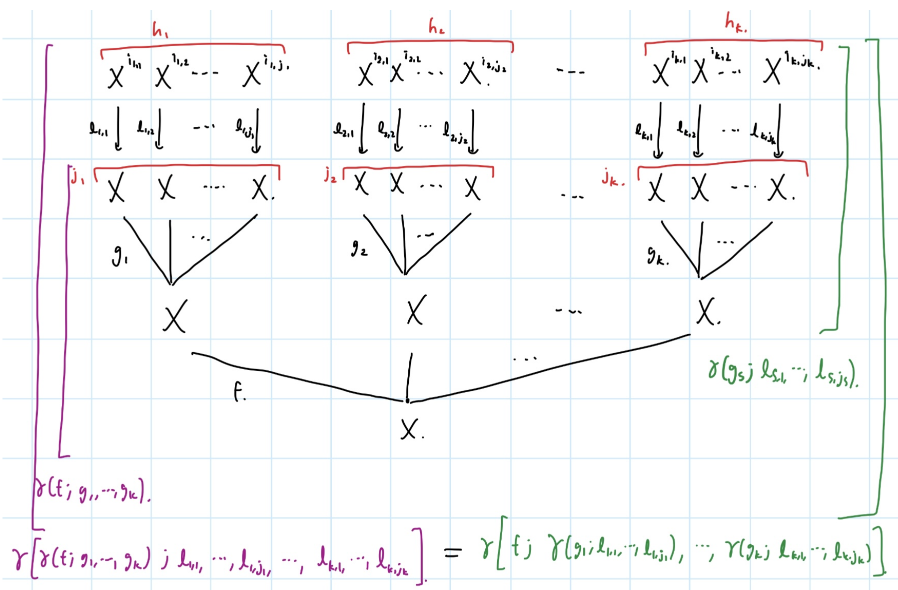
$$\gamma : \mathcal{E}_X(k) \times \prod_{i=1}^k \mathcal{E}_X(j_i) \to \mathcal{E}_X(j), \quad \gamma(f; g_1, ..., g_k) := f \circ \Pi g_i$$

There is also a canonical choice of *unit*, namely $\eta : \{*\} \to \mathcal{E}_X(1)$ by $* \mapsto \mathrm{id}_X$.

Unit here represents an *identity operation*, as one can verify the following two diagram commutes:

$$\begin{tikzcd}
\mathcal{E}_X(j) \times \{*\}^j \arrow[rr, "\sim"] \arrow[dr, "\mathrm{id}_{\mathcal{E}_X(j)} \times \eta^j"'] & & \mathcal{E}_X(j) \\
& \mathcal{E}_X(j) \times \mathcal{E}_X(1)^j \arrow[ur, "\gamma"'] &
\end{tikzcd}, \qquad
\begin{tikzcd}
\{*\} \times \mathcal{E}_X(j) \arrow[rr, "\sim"] \arrow[dr, "\eta \times \mathrm{id}_{\mathcal{E}_X(j)}"'] & & \mathcal{E}_X(j) \\
& \mathcal{E}_X(1) \times \mathcal{E}_X(j) \arrow[ur, "\gamma"'] &
\end{tikzcd}$$

It also satisfies a giant associativity law, for demonstration purpose I'll draw a tree diagram instead. Given k, $j = j_1 + ... + j_k$, each j_s associates with $h_s = i_{s,1} + ... + i_{s,j_s}$, and $i := h_1 + ... + h_k$. Then, the following combination diagram has no ambiguity, given $f \in \mathcal{E}_X(k)$, $g_s \in \mathcal{E}_X(j_s)$, and $\ell_{s,t} \in \mathcal{E}_X(i_{s,t})$:



Finally, there is a natural right Σ_j-action on $\mathcal{E}_X(j)$ by pre-permuting the input entries, $f \circ \sigma(x_1, ..., x_j) := f(x_{\sigma^{-1}(1)}, ..., x_{\sigma^{-1}(j)})$. Which, permutation in $\mathcal{E}_X(k)$ in γ is equivalent to "permuting orders" of blocks $(j_1, ..., j_k) = \{\{1, 2, ..., j_1\}, ..., \{j_1 + ... + j_{k-1} + 1, ..., j_1 + ... + j_{k-1} + j_k\}\}$ in $\{1, ..., j\}$, and each Σ_{j_s}-action on $\mathcal{E}_X(j_s)$ is equivalent to a permutation only on each block in $(j_1, ..., j_k)$.

Intuitively, each $\mathcal{E}_X(j)$ encodes all possible way of operating on j points of X (a j-ary operation on X), while each $\mathcal{E}_X(k)$ also encodes operation of combining k (possibly distinct) operations into another operation, in an associative way.

This is called the *Endomorphism Operad of* X. In general, if $\mathcal{V}$ is monoidally closed, the internal hom functor provides a construction of such endomorphism operad in $\mathcal{V}$.

Definition 2.2. Given symmetric monoidal category $\mathcal{V}$, a Σ *operad* $\mathcal{C}$ in $\mathcal{V}$ is a collection of objects $\{\mathcal{C}(j) \in \mathcal{V}\}_{j\in\mathbb{N}}$, each $\mathcal{C}(j)$ with a right Σ_j-action (a group homomorphism $\Sigma_j^{\mathrm{op}} \to \mathrm{Aut}_{\mathcal{V}}(\mathcal{C}(j))$), a *unit map* $\eta : 1 \to \mathcal{C}(1)$, together with composition maps

$$\gamma : \mathcal{C}(k) \otimes \mathcal{C}(j_1) \otimes ... \otimes \mathcal{C}(j_k) \to \mathcal{C}(j_1 + ... + j_k)$$

that satisfies the following axioms:

- *Unit:*

$$\begin{tikzcd}
\mathcal{C}(j) \otimes \kappa^{\otimes j} \arrow[rr, "\sim"] \arrow[dr, "\mathrm{id}_{\mathcal{C}(j)} \otimes \eta^{\otimes j}"'] & & \mathcal{C}(j) \\
& \mathcal{C}(j) \otimes \mathcal{C}(1)^{\otimes j} \arrow[ur, "\gamma"'] &
\end{tikzcd}, \qquad
\begin{tikzcd}
\kappa \otimes \mathcal{C}(j) \arrow[rr, "\sim"] \arrow[dr, "\eta \otimes \mathrm{id}_{\mathcal{C}(j)}"'] & & \mathcal{C}(j) \\
& \mathcal{C}(1) \otimes \mathcal{C}(j) \arrow[ur, "\gamma"'] &
\end{tikzcd}$$

- *Associativity:* Fix natural numbers k, $j = \sum_{s=1}^{k} j_s$, define index $g_s := j_1 + ... + j_s$ for $1 \leq s \leq k$. For each $i_1, ..., i_{g_k} = i_j$, with sum $i = \sum_{r=1}^{j} i_r$, and $h_s := i_{g_{(s-1)}+1} + ... + i_{g_s}$ (so $i = \sum_{s=1}^{k} h_s$ also), the following diagram commutes:

$$\begin{tikzcd}
\left[\mathcal{C}(k) \otimes \left(\bigotimes_{s=1}^{k} \mathcal{C}(j_s)\right)\right] \otimes \left(\bigotimes_{r=1}^{j} \mathcal{C}(i_r)\right) \arrow[r, "\gamma\otimes\mathrm{id}"] \arrow[dd, "\mathrm{shuffle}"'] & \mathcal{C}(j) \otimes \left(\bigotimes_{r=1}^{j} \mathcal{C}(i_r)\right) \arrow[dr, "\gamma"] & \\
& & \mathcal{C}(i) \\
\mathcal{C}(k) \otimes \left[\bigotimes_{s=1}^{k} \left(\mathcal{C}(j_s) \otimes \left(\bigotimes_{q=1}^{j_s} \mathcal{C}(i_{g_{(s-1)}+q})\right)\right)\right] \arrow[r, "\mathrm{id}_{\mathcal{C}(k)} \otimes (\bigotimes_s \gamma)"'] & \mathcal{C}(k) \otimes \left(\bigotimes_{s=1}^{k} \mathcal{C}(h_s)\right) \arrow[ur, "\gamma"'] &
\end{tikzcd}$$

- *Equivariance:*
  - Given $\sigma \in \Sigma_k$, $j_1 + ... + j_k = \Sigma_j$, let $\sigma(j_1, ..., j_k) \in \Sigma_j$ denotes "permutation of k-blocks of letters", by $(j_1, ..., j_k) \mapsto (j_{\sigma^{-1}(1)}, ..., j_{\sigma^{-1}(k)})$.

    Example 2.3. Given $j = 5 = 2 + 3$, the permutation $(1\ 2) \in \Sigma_2$, $\sigma(2, 3)$ is the permutation $\langle 1, 2; 3, 4, 5\rangle \mapsto \langle 3, 4, 5; 1, 2\rangle$ (namely changing the blocks' order, without changing the order in each block).

    Then, the following diagram commutes:

$$\begin{tikzcd}
\mathcal{C}(k) \otimes \left(\bigotimes_{r=1}^{k} \mathcal{C}(j_r)\right) \arrow[r, "\sigma\otimes\ \mathrm{shuffle}_{\sigma^{-1}}"] \arrow[d, "\gamma"'] & \mathcal{C}(k) \otimes \left(\bigotimes_{r=1}^{k} \mathcal{C}(j_{\sigma(r)})\right) \arrow[d, "\gamma"] \\
\mathcal{C}(j) \arrow[r, "\sigma(j_{\sigma(1)}, ..., j_{\sigma(k)})"'] & \mathcal{C}(j)
\end{tikzcd}$$

– Given $\tau_r \in \Sigma_{j_r}$, denote $\tau_1 \oplus ... \oplus \tau_k \in \Sigma_j$ by letting each τ_r permuting the j_rth block in j (for instance, let τ_1 acts on $1, ..., j_1$; let τ_2 acts on $j_1+1, ..., j_1+j_2$, etc.). Then, the following diagram commutes:

$$\begin{tikzcd} \mathcal{C}(k) \otimes \left(\bigotimes_{r=1}^k \mathcal{C}(j_r)\right) \arrow[r, "\mathrm{id}_{\mathcal{C}(k)} \otimes (\bigotimes_r \tau_r)"] \arrow[d, "\gamma"'] & \mathcal{C}(k) \otimes \left(\bigotimes_{r=1}^k \mathcal{C}(j_r)\right) \arrow[d, "\gamma"] \\ \mathcal{C}(j) \arrow[r, "\tau_1 \oplus ... \oplus \tau_k"'] & \mathcal{C}(j) \end{tikzcd}$$

If $\mathcal{C}(0) = \kappa$ (with $\gamma : \mathcal{C}(0) \to \mathcal{C}$ being $\mathrm{id}_{\mathcal{C}(0)}$), the operad is called *unital operad*. If omit the Σ_j-action, it's called *non-*Σ *operad*.

Definition 2.4. Let $\mathcal{C}, \mathcal{C}'$ be two operads on symmetric monoidal category $\mathcal{V}$, a *morphism of operads* $\psi : \mathcal{C} \to \mathcal{C}'$, is a sequence of Σ_j-equivariant morphisms $\psi_j : \mathcal{C}(j) \to \mathcal{C}'(j)$ in $\mathcal{V}$ (i.e., $\sigma^{-1} \circ \psi_j \circ \sigma = \psi_j$, for all $\sigma \in \Sigma_j$), such that it preserves the operad's structure morphism:

$$\begin{tikzcd} \mathcal{C}(k) \otimes \mathcal{C}(j_1) \otimes ... \otimes \mathcal{C}(j_k) \arrow[r, "\gamma"] \arrow[d, "\psi_k \otimes \psi_{j_1} \otimes ... \otimes \psi_{j_k}"'] & \mathcal{C}(j_1 + ... + j_k) \arrow[d, "\psi_{j_1 + ... + j_k}"] \\ \mathcal{C}'(k) \otimes \mathcal{C}'(j_1) \otimes .. \otimes \mathcal{C}'(j_k) \arrow[r, "\gamma'"'] & \mathcal{C}'(j_1 + ... + j_k) \end{tikzcd}$$

2.1 Categorical Example: Commutative Operad

Let $\mathcal{N}(j) := \kappa$, endows with trivial Σ_j-action. Define $\gamma : \mathcal{N}(k) \otimes \left(\bigotimes_{r=1}^k \mathcal{N}(j_r)\right) \to \mathcal{N}(j)$ to be the canonical isomorphism of unit. This is a simple (yet important definition wise) for defining E_∞-operad.

2.2 Categorical Example: Associative Operad

Let $\mathcal{M}(j) := \kappa[\Sigma_j] := \coprod_{\psi \in \Sigma_j} \kappa_\psi$, coproduct of $\#\Sigma_j$ copies of κ (each κ_ψ is the unit κ, subscript is for labeling convenience). It has a natural right Σ_j-action, where for each $\sigma \in \Sigma_j$, the identity $\kappa_{\psi\sigma^{-1}} \xrightarrow{\sim} \kappa_\psi$ induces an isomorphism on coproduct $\tilde{\sigma} \in \mathrm{Aut}(\mathcal{M}(j))$:

$$\begin{tikzcd} \kappa_\psi \arrow[r, "\sim"] \arrow[d, "\iota_\psi"'] & \kappa_{\psi\sigma} \arrow[d, "\iota_{\psi\sigma}"] \\ \coprod_{\psi \in \Sigma_j} \kappa_\psi \arrow[r, dashed, "\exists! \tilde{\sigma}"'] & \coprod_{\psi \in \Sigma_j} \kappa_{\psi\sigma} \end{tikzcd}$$

This construction enforces $(\widetilde{\sigma\sigma'}) = \tilde{\sigma}' \circ \tilde{\sigma}$, which is a group homomorphism $\Sigma_j^{\mathrm{op}} \to \mathrm{Aut}_{\mathcal{V}}(\mathcal{M}(j))$.

There is a natural unit morphism $\eta : \kappa \to \kappa[\Sigma_1] = \kappa$ via identity.

Also, each identity morphism $\kappa_\psi \to \kappa$ induces a trivialization $t : \kappa[\Sigma_j] \to \kappa$. Combining the action $\psi \in \Sigma_k$ (by identifying concatination of blocks $(j_{\psi^{-1}(1)}, ..., j_{\psi^{-1}(k)}) = \{1, ..., j\}$ in order, also inducing inclusion $\Sigma_{j_s} \hookrightarrow \Sigma_j$), we get the following morphism:

$$\kappa_\psi \otimes \left(\bigotimes_{s=1}^k \kappa[\Sigma_{j_s}]\right) \xrightarrow{\sim} \bigotimes_{s=1}^k \kappa[\Sigma_{j_{\psi^{-1}(s)}}] \xrightarrow{t^{k-1}} \kappa[\Sigma_{j_{\psi^{-1}(s)}}] \to \kappa[\Sigma_j]$$

Running through all $\psi \in \Sigma_k$, this induces a morphism $\gamma : \coprod_{\psi\in\Sigma_k} \left[\kappa_\psi \otimes \left(\bigotimes_{s=1}^k \kappa[\Sigma_{j_s}]\right)\right] \to \kappa[\Sigma_j]$, and use the fact $\otimes$ commutes with colimits, this is equivalent to a morphism $\gamma : \kappa[\Sigma_k] \otimes \left(\bigotimes_{s=1}^k \kappa[\Sigma_{j_s}]\right) \to \kappa[\Sigma_j]$.

Verify $\mathcal{M}$ forms an operad is a long tedious work, unless many people request the proof we'll omit it here :)

As a side note, this is also crucial for defining A_∞-operad.

2.3 Topological Example: Stasheff Associahedra (To do)

Given a based space X, its loop space $Y = \Omega X := F(S^1, X)$ has a product via concatination: $Y^2 \to Y$ by $(g, f) \mapsto g \cdot f$, where

$$(g \cdot f)(t) = \begin{cases} f(2t) & 0 \leq t \leq \frac{1}{2} \\ g(2t-1) & \frac{1}{2} < t \leq 1 \end{cases}$$

when viewing $S^1 = I/\partial I$.

3 Operad Algebra

As a motivating example, let's start with the endomorphism operad:

Example 3.1. Take $X \in \mathcal{T}^*$ and its endomorphism operad, there is a collection of canonical maps $\theta : \mathcal{E}_X(j) \times X^j \to X$, given by $\theta(f; x_1, ..., x_j) := f(x_1, ..., x_j)$.

It inherits the associativity, because the resulting action on X^j disregards if we plugin the entries first then combine the multimorphisms, or combine the multimorphisms then plugin the entries.

The identity action reflects its name, as $\{*\} \times X \to \mathcal{E}_X(1) \times X \to X$ by $(*; x) \mapsto (id_X; x) \mapsto \mathrm{id}_X(x) = x$ is an isomorphism.

Also, notice each X^j has a canonical left Σ_j-action by permutation, $\sigma(x_1, ..., x_j) = (x_{\sigma(1)}, ..., x_{\sigma(j)})$, and notice that the Σ_j-action on $\mathcal{E}_X(j) \times X^j$ by $(f\sigma; (x_1, ..., x_j)) = (f; \sigma^{-1}(x_1, ..., x_j))$, so the Σ_j-action can be communicated between the product's entries.

Definition 3.2. Given an operad $\mathcal{C}$ over a symmetric monoidal category $\mathcal{V}$, a *$\mathcal{C}$-algebra* is an object $X \in \mathcal{V}$, with collections of morphisms $\theta : \mathcal{C}(j) \otimes X^j \to X$, that satisfies the following axioms:

- *Associativity:* Given $j = \sum_{i=1}^k j_i$, the following diagram commutes:

$$\begin{tikzcd}
\left[C(k) \otimes \left(\bigotimes_{i=1}^k C(j_i)\right)\right] \otimes A^{\otimes j} \arrow[r, "\gamma\otimes \mathrm{id}_{A^{\otimes j}}"] \arrow[dd, "\text{shuffle}"'] & C(j) \otimes A^{\otimes j} \arrow[dr, "\theta"] & \\
 & & A \\
C(k) \otimes \left[\bigotimes_{i=1}^k \left(C(j_i) \otimes A^{\otimes j_i}\right)\right] \arrow[r, "\mathrm{id}_{C(k)}\otimes\theta^k"'] & C(k) \otimes A^{\otimes k} \arrow[ur, "\theta"'] &
\end{tikzcd}$$

  (Note: This part uses the isomorphism $A^{\otimes j} \cong \bigotimes_{i=1}^k A^{\otimes j_i}$).

- *Unit:* The following diagram commutes:

$$\begin{tikzcd}
\kappa \otimes A \arrow[rr, "\sim"] \arrow[dr, "\eta\otimes\mathrm{id}_A"'] & & A \\
 & C(1) \otimes A \arrow[ur, "\theta"'] &
\end{tikzcd}$$

- *Eauivariance:* Given any permutation $\sigma \in \Sigma_j$, the following diagram commutes:

$$\begin{tikzcd}
C(j) \otimes A^{\otimes j} \arrow[rr, "\sigma\otimes\ \text{shuffle}_\sigma"] \arrow[dr, "\theta"'] & & C(j) \otimes A^{\otimes j} \arrow[dl, "\theta"] \\
 & A &
\end{tikzcd}$$

For instance, any loop space is automatically a $\mathcal{K}$-space, acted by the Stasheff Associahedras.

Example 3.3. A $\mathcal{N}$-algebra $A \in \mathcal{V}$ is a *commutative monoid* in the symmetric monoidal category: It has a product morphism $\theta : \mathcal{N}(2) \otimes A^2 \cong A \otimes A \to A$, and a unit morphism $\theta \circ \eta : A \cong \kappa \otimes A \xrightarrow{\sim} \mathcal{N}(1) \otimes A \xrightarrow{\sim} A$.

- It's associative based on the following diagram and something analogous:

$$\begin{tikzcd}
\mathcal{N}(2)\otimes\mathcal{N}(1)\otimes\mathcal{N}(2)\otimes A^3 \arrow[rr, "\gamma\otimes\mathrm{id}_{A^3}"] \arrow[d, "\text{shuffle}"'] & & \mathcal{N}(3)\otimes A^3 \arrow[dr, "\theta"] & \\
& & & A \\
\mathcal{N}(2)\otimes(\mathcal{N}(1)\otimes A)\otimes\big(\mathcal{N}(2)\otimes A^2\big) \arrow[rr, "\mathrm{id}_{\mathcal{N}(2)}\otimes\theta^2"'] & & \mathcal{N}(2)\otimes A^2 \arrow[ur, "\theta"'] &
\end{tikzcd}$$

This diagram is formally saying the 3-ary multiplication is equivalent to "multiply" the last two entries, then multiply with the first entry. Interchange the 1 and 2, we get the other order of multiplication.

- It's commutative because taking transposition $\tau \in \Sigma_2$ (with trivial action on $\mathcal{N}(2)$), the equivariance diagram provides:

$$\begin{tikzcd}
\mathcal{N}(2)\otimes(A_{(1)}\otimes A_{(2)}) \arrow[rr, "\tau\otimes\ \text{shuffle}_\tau"] \arrow[dr, "\theta"'] & & \mathcal{N}(2)\otimes(A_{(2)}\otimes A_{(1)}) \arrow[dl, "\theta"] \\
& A &
\end{tikzcd}$$